GREGORY GOLONKA

gregorygolonka@outlook.com | 802-595-2750 | linkedin.com/in/gregory-golonka-b99a61245/ | https://gtgolonka.com/

Education

University of Notre Dame

- B.S. Aerospace Engineering, May 2025
- GPA: 3.53
- Relevant Coursework: Intermediate Thermodynamics, Fluid Mechanics, Differential Equations I & II, Heat Transfer, Theoretical Aerodynamics, Compressible Aerodynamics, Aerospace Structures, Aerospace Dynamics, Gas Turbines and Propulsions, Flight Mechanics and Intro to Design, Mechanisms and Machines, Orbital and Space Mechanics, Senior Design

Technical Skills

- Technical & Testing Skills: Mechanical testing, materials characterization, load frame setup, data acquisition, strain analysis, LabVIEW (exposure)
- Programming & Simulation: Python, MATLAB, Fortran, C++, HTML, Next.js, LabView (exposure)
- CAD & Design: SolidWorks, Autodesk Fusion 360, XFLR5, XFoil
- Tools: CHEMKIN-II, LSODE, ONELAB/Gmsh, Wind Tunnel Instrumentation
- Engineering: Propulsion Systems, Flight Dynamics, Experimental Aerodynamics
- Soft Skills: Technical documentation, Leadership, Project Management, Problem Solving, Independent Learning, Collaboration, Adaptability and Flexibility, Communication, Work Ethic, FMEA, RCCA

Projects & Research

Senior Design — Radio-Controlled Aircraft **2024 - 2025**

- Designed a radio-controlled airplane to complete a specific mission, focusing on aerodynamic optimization and structural integrity using finite element analysis (FEA).
- Utilized XFLR5 and SOLIDWORKS CFD to analyze and refine the base model through iterative simulations.
- Led budgeting, logistics, and construction; managed CAD-guided assembly and sourced materials, balancing cost, weight, and durability.

Undergraduate Research — Shock Wave Structure Analysis **2024 - 2025**

- Conducted independent study into viscous shock structures under Dr. Joseph M. Powers.
- Investigated dissociation of diatomic molecules under extreme shock conditions using 1D Navier-Stokes equations.
- Advanced understanding of propulsion and detonation modeling by linking molecular dynamics to continuum-scale physics.

Computing Project — Heat Transfer in Food Systems **2025**

- Developed a numerical model to simulate transient heat conduction in a steak during cooking.
- Accounted for variable oven temperatures to optimize char formation while maintaining sub-medium-rare interior.
- Created predictive model for cook-time estimation under uncertain parameters.

Personal Project — Portfolio Website -- https://gtgolonka.com/ **2024 - 2025**

- Built first personal website using JavaScript, HTML, Next.js, and CSS.
- Strengthened skills in frontend development and project self-learning.

Gas Turbines & Propulsion — Williams F-107 Engine Redesign **2025**

- Designed and optimized MATLAB-based mission simulations for the BGM-109A Tomahawk, analyzing thrust, fuel, and range.
- Conducted engine performance studies and sensitivity analysis, achieving improved efficiency and extended standoff distance

Experimental Aerodynamics — Wind Tunnel Studies **2024**

- Calibrated pressure transducers and collected digital data from analog signals.
- Analyzed airfoil circulation, pressure distribution, and lift/drag characteristics of finite wings.
- Developed strong foundation in experimental methods and aerodynamic force analysis.

Work Experience

Movie Theater Projectionist — Capitol Theater **2023-Present**

- Troubleshoot and maintain digital projection systems to ensure uptime of events. Coordinate employee scheduling and track inventory management. Record financial spending and statements using QuickBooks.

Dorm Front Desk Worker— Dunne Hall, University of Notre Dame **2022-2023**

- Managed front desk operations and supported in-dorm services, ensuring resident needs and safety were met.

Landscape Technician — C. Michaud Landscaping **2021-2024**

- Coordinated team operations across 150+ weekly properties, maintaining high service quality.